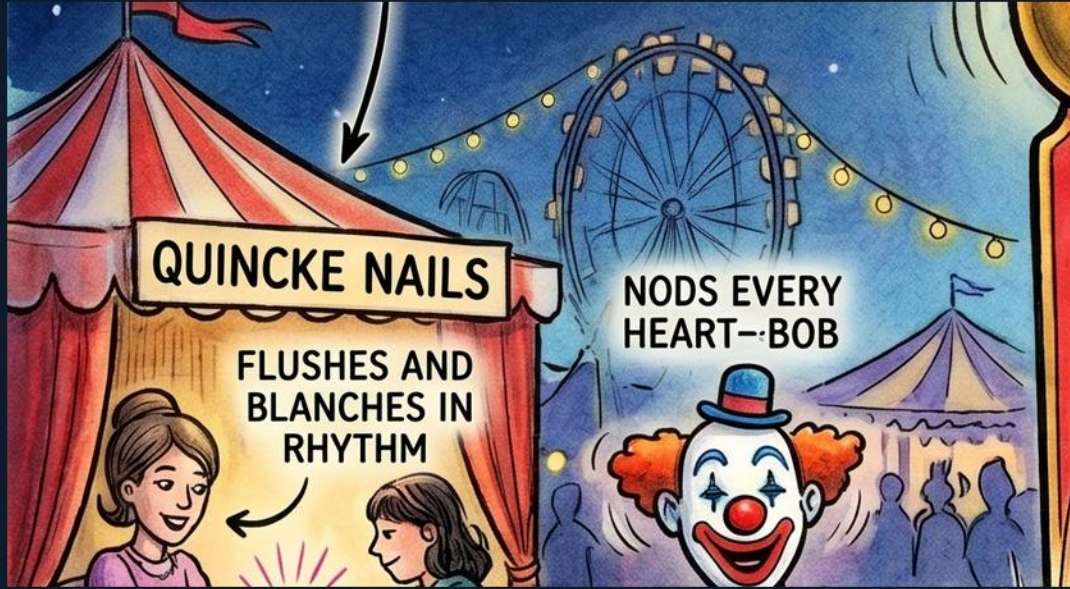


Valvular — Aortic Regurgitation





Quincke nail-bed pulsations (booth)

WHAT YOU SEE

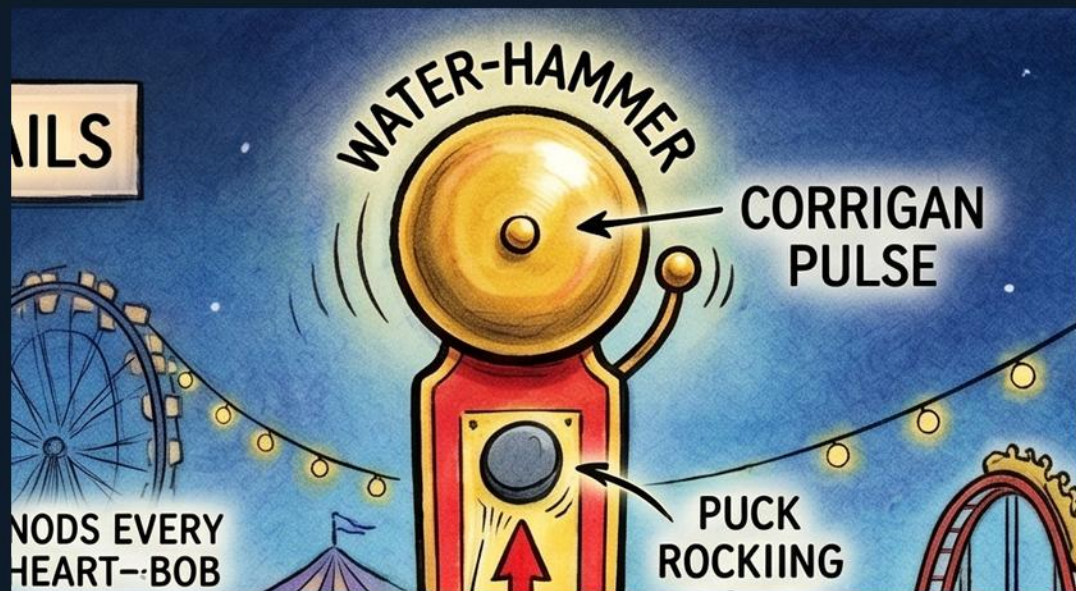
A fairground booth labeled QUINCKE NAILS — FLUSHES AND BLANCHES IN RHYTHM.

WHAT IT ENCODES

Quincke's sign is visible capillary pulsation in the nail bed — the nail blanches and flushes with each heartbeat — driven by the wide pulse pressure of chronic aortic regurgitation. It is one of the eponymous peripheral signs of a hyperdynamic circulation. Seen by lightly compressing the nail.

THE BOARD TRAP

Quincke pulsations reflect WIDE PULSE PRESSURE (chronic AR), not capillary refill or anemia — don't attribute them to perfusion problems.



Water-hammer / Corrigan pulse (bell + mallet)

WHAT YOU SEE

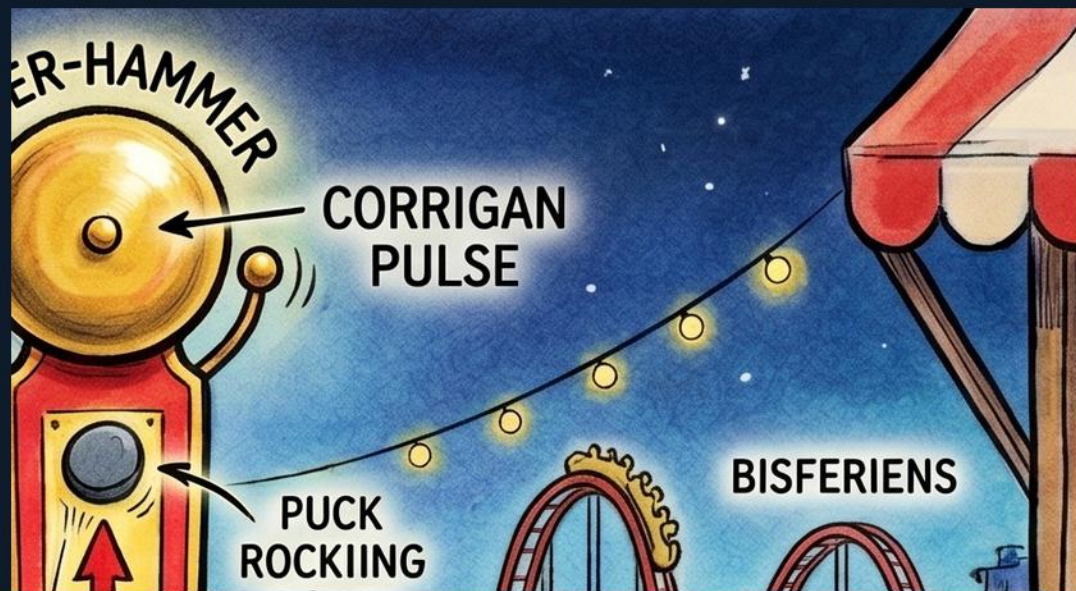
A large bell struck by a mallet labeled WATER-HAMMER and CORRIGAN PULSE.

WHAT IT ENCODES

The water-hammer (Corrigan) pulse is a rapidly rising and rapidly collapsing arterial pulse from the large stroke volume and diastolic runoff of AR. It is best appreciated by raising the patient's arm. The bounding-then-collapsing quality is the hallmark peripheral sign.

THE BOARD TRAP

A bounding, collapsing pulse is AR — opposite of the weak, slow-rising parvus et tardus of aortic STENOSIS; don't confuse the two.



Corrigan pulse label

WHAT YOU SEE

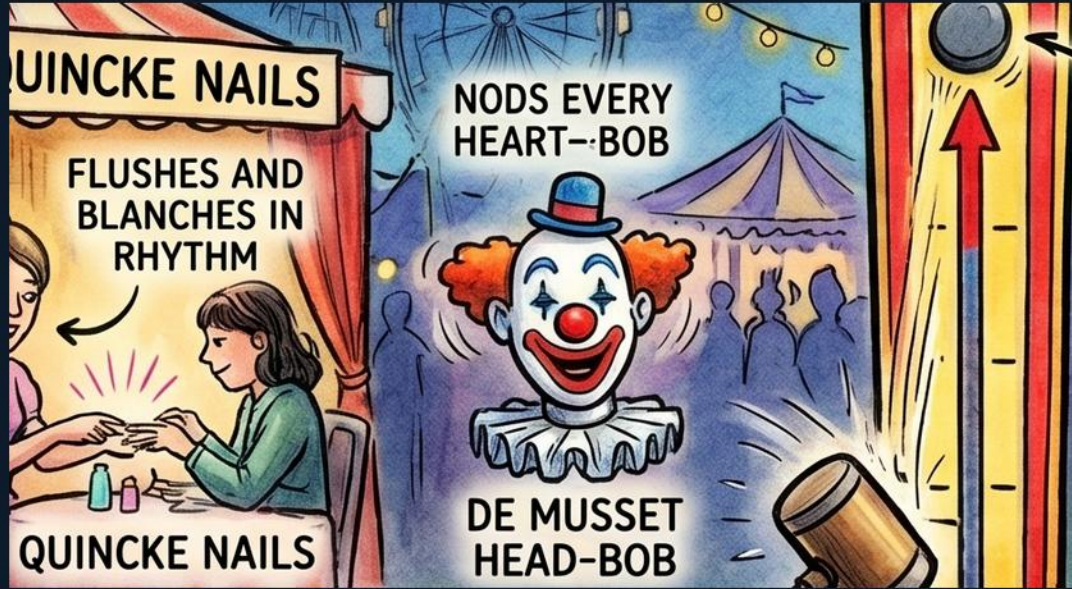
Text CORRIGAN PULSE beside the water-hammer bell.

WHAT IT ENCODES

Corrigan's pulse is the carotid manifestation of the same water-hammer phenomenon — a visible, abrupt, full upstroke followed by quick collapse. Corrigan and water-hammer name the same finding. It signals significant chronic AR with wide pulse pressure.

THE BOARD TRAP

Corrigan (visible carotid) and water-hammer (palpated radial) are the SAME hemodynamic sign of AR — not two different lesions.



De Musset head-bob (clown)

WHAT YOU SEE

A clown **NODS EVERY HEART-BEAT**, labeled **DE MUSSET HEAD-BOB**.

WHAT IT ENCODES

De Musset's sign is rhythmic bobbing of the head with each heartbeat, caused by the forceful systolic ejection and wide pulse pressure of AR transmitted to the carotids. It is a classic visible eponym. Like the other peripheral signs, it reflects a large stroke volume.

THE BOARD TRAP

Head-bobbing is De Musset's sign of AR (wide pulse pressure) — not a neurologic tremor or movement disorder.



Pulsus bisferiens (roller-coaster)

WHAT YOU SEE

A double-humped roller-coaster labeled BISFERIENS.

WHAT IT ENCODES

Pulsus bisferiens is a pulse with TWO systolic peaks per beat, classically felt in significant AR (and in mixed AS/AR or HOCM). The double hump reflects rapid ejection of a large stroke volume. It is another peripheral marker of the regurgitant load.

THE BOARD TRAP

Bisferiens = two peaks in SYSTOLE (AR or mixed AS/AR); don't confuse it with the dicrotic pulse (a systolic + diastolic peak in low-output states).



Pistol-shot femorals (gunslinger)

WHAT YOU SEE

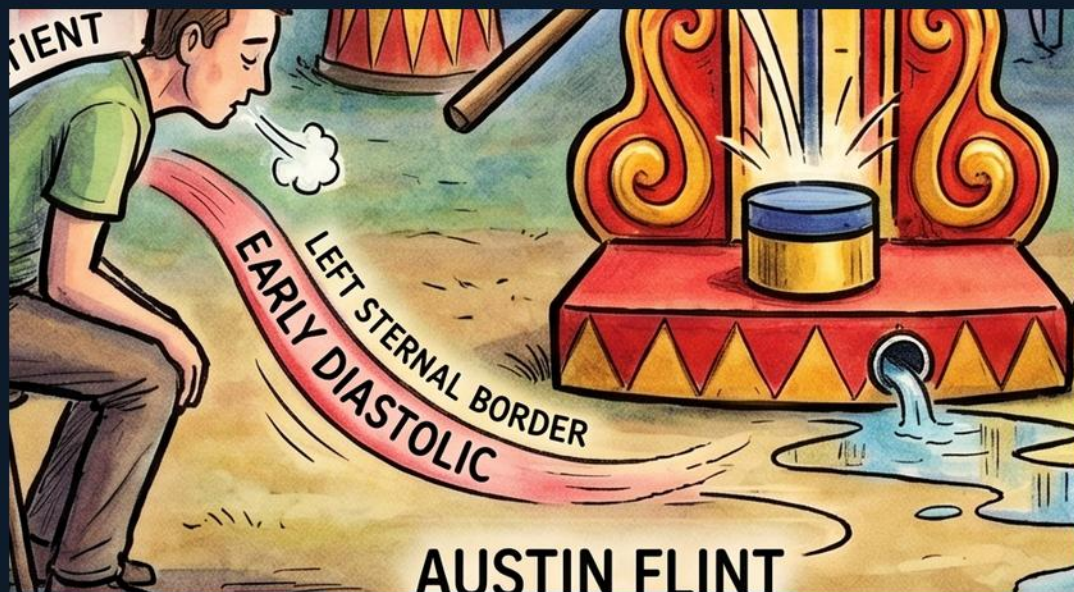
A figure firing a pistol with a LITTLE BANG, labeled PISTOL-SHOT FEMORALS.

WHAT IT ENCODES

Traube's sign / pistol-shot femorals are loud, sharp, double sounds auscultated over the femoral arteries in AR. They arise from the abrupt distension and collapse of the artery with the wide pulse pressure. Another auscultatory peripheral sign of severe AR.

THE BOARD TRAP

Pistol-shot femorals are a sign of wide-pulse-pressure AR — not a femoral bruit of arterial stenosis/atherosclerosis.



Left sternal border — early diastolic murmur

WHAT YOU SEE

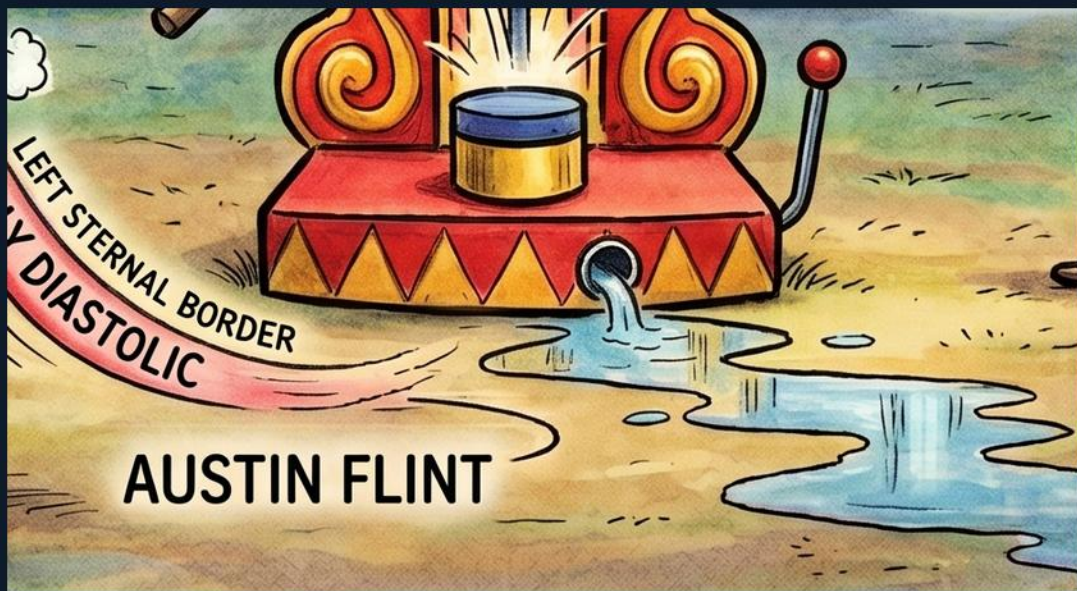
A patient leaning forward labeled LEAN FORWARD EXPIRE / LEFT STERNAL BORDER / EARLY DIASTOLIC.

WHAT IT ENCODES

The AR murmur is an EARLY DIASTOLIC decrescendo murmur heard best at the LEFT sternal border with the patient sitting up, leaning forward, in held EXPIRATION. These maneuvers bring the aortic root closer to the chest wall. Its early-diastolic decrescendo shape and position define it.

THE BOARD TRAP

The AR murmur is best heard sitting forward in EXPIRATION at the left sternal border — not lying flat; missing the maneuver misses the murmur.



Austin Flint murmur (apex rumble)

WHAT YOU SEE

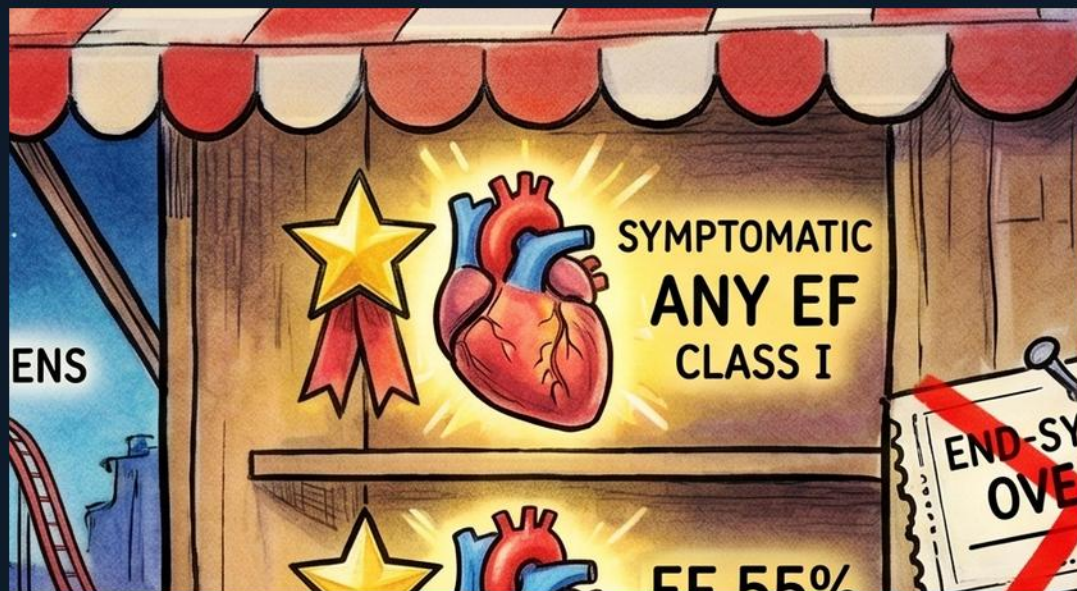
Text AUSTIN FLINT at the base, near the apex region.

WHAT IT ENCODES

The Austin Flint murmur is a low-pitched mid-to-late DIASTOLIC rumble at the APEX in AR, produced as the regurgitant jet impinges on the anterior mitral leaflet, causing functional mitral inflow obstruction. It can mimic mitral stenosis. There is no opening snap and no loud S1, which helps distinguish it.

THE BOARD TRAP

An apical diastolic rumble in AR is Austin Flint (functional) — don't diagnose true mitral stenosis; there's no opening snap or loud S1.



Symptomatic any EF = Class I (top ribbon)

WHAT YOU SEE

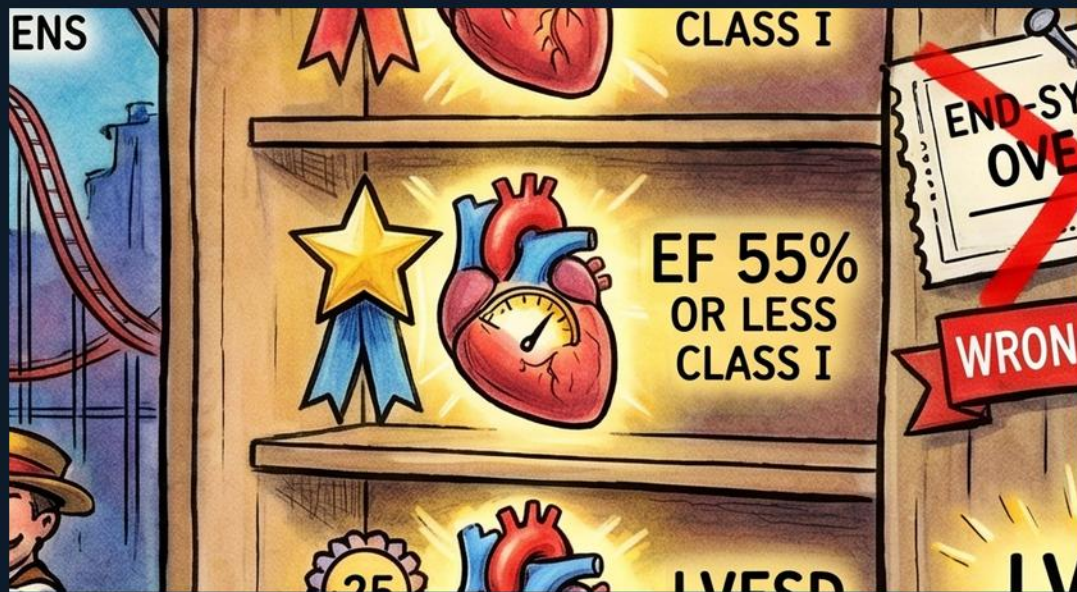
A vendor shelf ribbon: SYMPTOMATIC ANY EF — CLASS I, with a heart.

WHAT IT ENCODES

For severe AR, symptoms are an immediate Class I indication for surgery REGARDLESS of EF — exertional dyspnea, angina, or heart failure should not be tolerated. Waiting risks irreversible LV dysfunction. Symptom onset = operate.

THE BOARD TRAP

Operate on symptomatic severe AR at ANY EF — a preserved EF does not justify delaying surgery once the patient is symptomatic.



EF 55% or less = Class I (middle ribbon)

WHAT YOU SEE

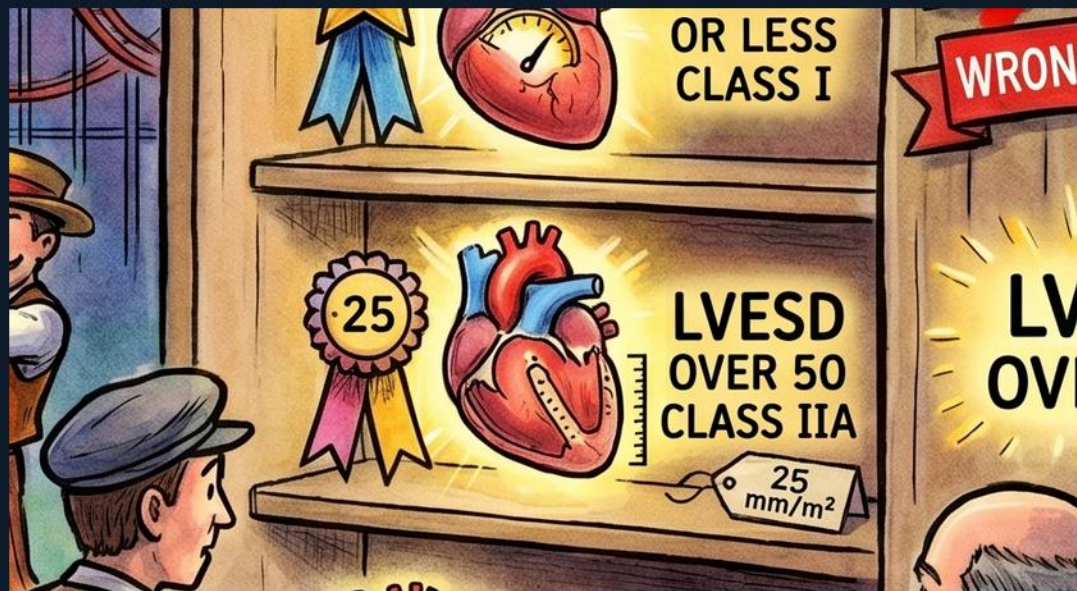
Second ribbon: EF 55% OR LESS — CLASS I, with a heart.

WHAT IT ENCODES

In ASYMPTOMATIC severe AR, an LVEF $\leq 55\%$ is a Class I indication for surgery. Declining systolic function signals the ventricle is failing under the volume load even before symptoms appear. The 55% threshold is the asymptomatic EF cutoff.

THE BOARD TRAP

The asymptomatic-AR EF cutoff is $\leq 55\%$ (Class I) — using 50% (the AS number) here is wrong.



LVESD over 50 = Class IIa (lower ribbon)

WHAT YOU SEE

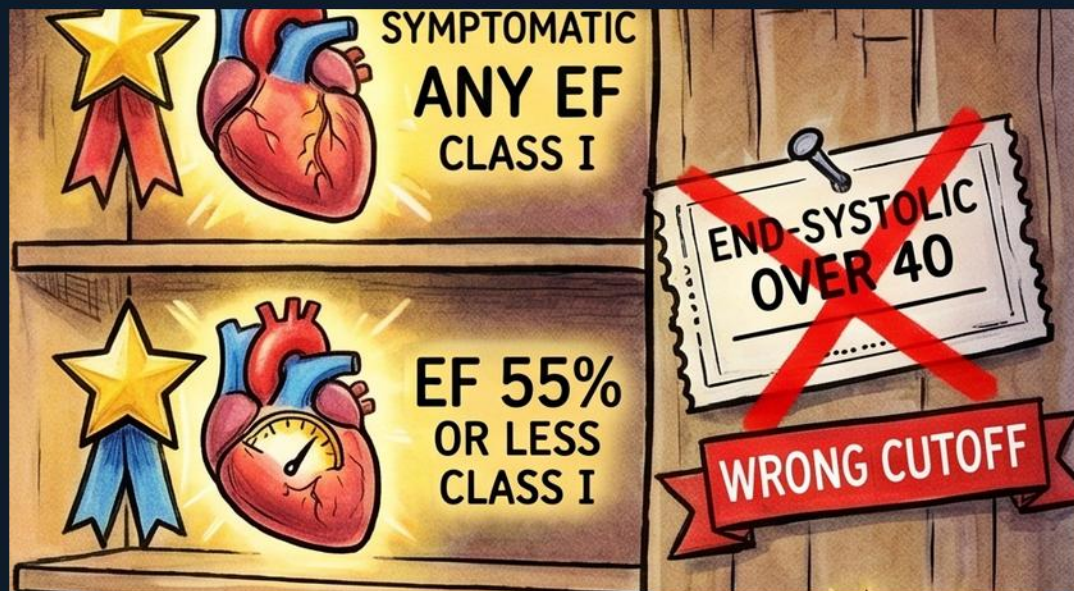
Third ribbon: LVESD OVER 50 — CLASS IIa, with 25 mm/m² cue.

WHAT IT ENCODES

In asymptomatic severe AR with EF >55%, surgery is Class IIa when the LV is dilating: LVESD >50 mm (or indexed >25 mm/m²). Progressive end-systolic enlargement predicts adverse remodeling. This is the dimension-based trigger for earlier intervention.

THE BOARD TRAP

The dimension trigger is LVESD >50 mm (or >25 mm/m² indexed) — NOT 40 mm; the '>40 mm end-systolic' cutoff is the classic wrong answer.



End-systolic over 40 = WRONG CUTOFF (red X)

WHAT YOU SEE

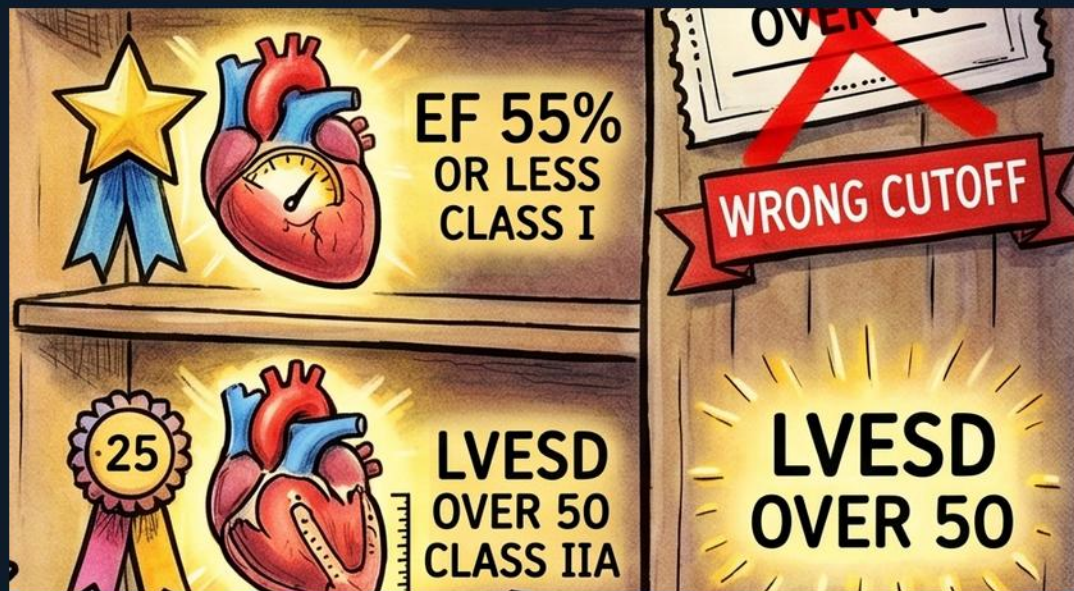
A red X over a sign reading END-SYSTOLIC OVER 40 — WRONG CUTOFF.

WHAT IT ENCODES

The 'end-systolic dimension >40 mm' figure is a deliberate distractor — the real surgical dimension cutoff in asymptomatic severe AR is LVESD >50 mm (or >25 mm/m²). Anchoring to 40 mm leads to operating too early on the wrong basis. Reject this number.

THE BOARD TRAP

LVESD >40 mm is the WRONG cutoff for AR surgery — the correct value is >50 mm / >25 mm/m²; this is the planted board trap.



LVESD OVER 50 (correct, plain sign)

WHAT YOU SEE

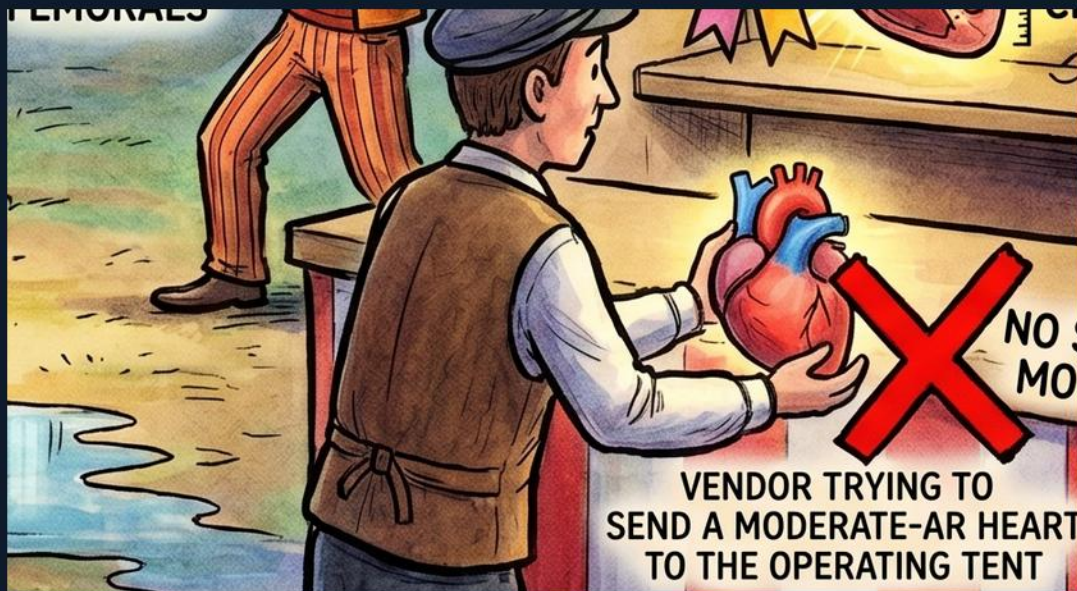
A plain sign reading LVESD OVER 50 (no X), the correct value highlighted.

WHAT IT ENCODES

This restates the CORRECT dimension threshold: LVESD >50 mm (or >25 mm/m² indexed) drives Class IIa surgery in asymptomatic severe AR with preserved EF. Keep the right number front of mind against the 40 mm distractor. It pairs with the EF $\leq 55\%$ Class I rule.

THE BOARD TRAP

Memorize LVESD >50 mm / >25 mm/m² as the AR dimension cutoff — the value with the red X (>40) is there to mislead.



No surgery for moderate AR (vendor + red X)

WHAT YOU SEE

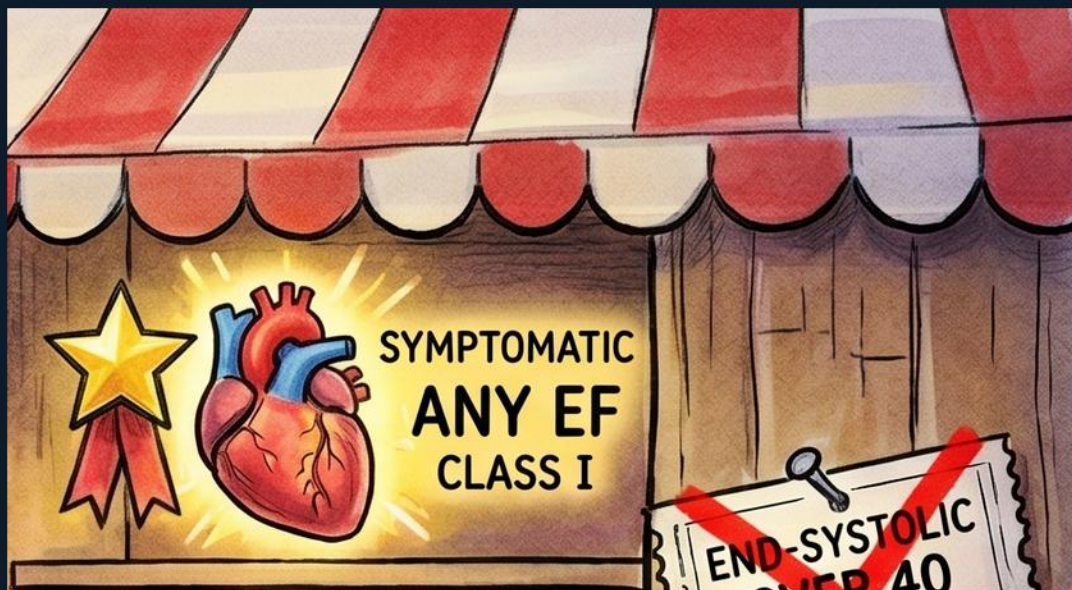
A vendor pushing a MODERATE heart to the operating tent, blocked by a red X labeled NO SURGERY MODERATE.

WHAT IT ENCODES

These surgical thresholds apply only to SEVERE AR. MODERATE AR is not an indication for valve surgery on its own — it is managed medically with surveillance. Don't be lured into operating on a moderate lesion. Severity must be established before applying the Class I/IIa rules.

THE BOARD TRAP

Don't operate on MODERATE AR — the symptom/EF/LVESD criteria apply to SEVERE AR only; sending a moderate heart to surgery is the trap.



Carnival/fairground setting (wide pulse pressure theme)

WHAT YOU SEE

A festive fairground backdrop tying the bouncing, bobbing, banging signs together.

WHAT IT ENCODES

The whole carnival theme encodes the hyperdynamic, WIDE PULSE PRESSURE circulation of chronic AR — every bouncing, bobbing, and banging eponym (Corrigan, De Musset, Quincke, Traube, bisferiens) stems from a large stroke volume with rapid diastolic runoff. Wide pulse pressure is the unifying mechanism. The peripheral signs are all manifestations of it.

THE BOARD TRAP

All the eponymous peripheral signs share ONE cause — wide pulse pressure of chronic AR; they are not independent findings to memorize in isolation.

Summary — every symbol

1. Quincke nail-bed pulsations (booth)

Quincke's sign is visible capillary pulsation in the nail bed — the nail blanches and flushes with each heartbeat — driven by the wide pulse pressure of chronic aortic regurgitation

3. Corrigan pulse label

Corrigan's pulse is the carotid manifestation of the same water-hammer phenomenon — a visible, abrupt, full upstroke followed by quick collapse

5. Pulsus bisferiens (roller-coaster)

Pulsus bisferiens is a pulse with TWO systolic peaks per beat, classically felt in significant AR (and in mixed AS/AR or HOCM)

7. Left sternal border — early diastolic murmur

The AR murmur is an EARLY DIASTOLIC decrescendo murmur heard best at the LEFT sternal border with the patient sitting up, leaning forward, in held EXPIRATION

9. Symptomatic any EF = Class I (top ribbon)

For severe AR, symptoms are an immediate Class I indication for surgery REGARDLESS of EF — exertional dyspnea, angina, or heart failure should not be tolerated

11. LVESD over 50 = Class IIa (lower ribbon)

In asymptomatic severe AR with EF >55%, surgery is Class IIa when the LV is dilating: LVESD >50 mm (or indexed >25 mm/m²)

13. LVESD OVER 50 (correct, plain sign)

This restates the CORRECT dimension threshold: LVESD >50 mm (or >25 mm/m² indexed) drives Class IIa surgery in asymptomatic severe AR with preserved EF

15. Carnival/fairground setting (wide pulse pressure theme)

The whole carnival theme encodes the hyperdynamic, WIDE PULSE PRESSURE circulation of chronic AR — every bouncing, bobbing, and banging eponym (Corrigan, De Musset, Quincke, Traube, bisferiens) stems from a large stroke volume with rapid diastolic runoff

2. Water-hammer / Corrigan pulse (bell + mallet)

The water-hammer (Corrigan) pulse is a rapidly rising and rapidly collapsing arterial pulse from the large stroke volume and diastolic runoff of AR

4. De Musset head-bob (clown)

De Musset's sign is rhythmic bobbing of the head with each heartbeat, caused by the forceful systolic ejection and wide pulse pressure of AR transmitted to the carotids

6. Pistol-shot femorals (gunslinger)

Traube's sign / pistol-shot femorals are loud, sharp, double sounds auscultated over the femoral arteries in AR

8. Austin Flint murmur (apex rumble)

The Austin Flint murmur is a low-pitched mid-to-late DIASTOLIC rumble at the APEX in AR, produced as the regurgitant jet impinges on the anterior mitral leaflet, causing functional mitral inflow obstruction

10. EF 55% or less = Class I (middle ribbon)

In ASYMPTOMATIC severe AR, an LVEF ≤55% is a Class I indication for surgery

12. End-systolic over 40 = WRONG CUTOFF (red X)

The 'end-systolic dimension >40 mm' figure is a deliberate distractor — the real surgical dimension cutoff in asymptomatic severe AR is LVESD >50 mm (or >25 mm/m²)

14. No surgery for moderate AR (vendor + red X)

These surgical thresholds apply only to SEVERE AR